

Data Cleaning and Preliminary Analysis in R

Case study: Mammal Sleep Dataset (msleep)

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Table of Contents

1. Introduction and Objective 2
2. Dataset Overview 3
2.1 *Source and rationale* 3
2.2 *Loading the data* 3
2.3 *Data dictionary* 3
3. Initial Inspection 4
3.1 *Structure* 4
3.2 *Summary statistics of the raw data* 5
3.3 *Missing values* 6
3.4 *Duplicates* 7
4. Data Cleaning and Transformation 7
4.1 *Type correction and text standardisation* 7
4.2 *Handling missing values* 8
4.3 *Outlier detection* 10
4.4 *Transformation and normalisation* 12
4.5 *Encoding categorical variables* 13
4.6 *Cleaned dataset* 15
5. Exploratory Data Analysis 15
5.1 *Structure and summary of the cleaned data* 15
5.2 *Descriptive statistics* 16
5.3 *Distributions* 17
5.4 *Relationships between variables* 19
5.5 *Correlation analysis* 21
6. Key Insights 22
7. Conclusion and Limitations 23
8. Screenshots of the RStudio Workflow 23
Appendix: Reproducibility 24

1. Introduction and Objective

This report documents the data cleaning, preprocessing and preliminary analysis of a publicly available dataset using R.

The objectives are to:

1. Select a dataset with missing values and a mix of categorical and numerical variables.
2. *Clean it:* handle missing values, detect outliers, normalise numeric variables and encode categorical variables.
3. Carry out exploratory data analysis (EDA) with str(), summary(), descriptive statistics, correlations and visualisations.
4. Record the reasoning behind every decision so the workflow is reproducible.

2. Dataset Overview

2.1 *Source and rationale*

The **msleep** dataset ships with the `ggplot2` package. It is an updated and expanded version of the mammal sleep data of Allison and Cicchetti (1976), and contains sleep patterns, brain weight and body weight for 83 mammal species. It was chosen because it has:

- **Missing values** in both numeric and categorical columns;
- **A mix of variable types** (character/categorical and numeric);
- **Strong outliers and skew** (e.g. elephants versus mice in body weight), which makes normalisation and outlier handling meaningful rather than cosmetic.

2.2 *Loading the data*

```
data("msleep", package = "ggplot2")
raw <- msleep          # keep an untouched copy for before/after comparisons
dim(raw)

## [1] 83 11
```

2.3 *Data dictionary*

```
dictionary <- tibble(
  Variable = c("name", "genus", "vore", "order", "conservation",
               "sleep_total", "sleep_rem", "sleep_cycle", "awake",
               "brainwt", "bodywt"),
  Type = c(rep("Categorical", 5), rep("Numeric", 6)),
  Description = c("Common name of the animal", "Genus", "Diet: carni, herbi,
insecti, omni",
                  "Taxonomic order", "Conservation status (IUCN-style code)",
                  "Total sleep per day (hours)", "REM sleep per day (hours)",
                  "Length of sleep cycle (hours)", "Hours awake per day",
                  "Brain weight (kg)", "Body weight (kg)")
)
kable(dictionary)
```

Variable	Type	Description
name	Categorical	Common name of the animal
genus	Categorical	Genus
vore	Categorical	Diet: carni, herbi, insecti, omni
order	Categorical	Taxonomic order
conservation	Categorical	Conservation status (IUCN-style code)
sleep_total	Numeric	Total sleep per day (hours)
sleep_rem	Numeric	REM sleep per day (hours)
sleep_cycle	Numeric	Length of sleep cycle (hours)
awake	Numeric	Hours awake per day
brainwt	Numeric	Brain weight (kg)
bodywt	Numeric	Body weight (kg)

3. Initial Inspection

3.1 *Structure*

```
str(raw)
```

```
## tibble [83 × 11] (S3: tbl_df/tbl/data.frame)
##  $ name        : chr [1:83] "Cheetah" "Owl monkey" "Mountain beaver" "Grea
ter short-tailed shrew" ...
##  $ genus       : chr [1:83] "Acinonyx" "Aotus" "Aplodontia" "Blarina" ...
##  $ vore        : chr [1:83] "carni" "omni" "herbi" "omni" ...
##  $ order       : chr [1:83] "Carnivora" "Primates" "Rodentia" "Soricomorph
a" ...
##  $ conservation: chr [1:83] "lc" NA "nt" "lc" ...
##  $ sleep_total : num [1:83] 12.1 17 14.4 14.9 4 14.4 8.7 7 10.1 3 ...
##  $ sleep_rem   : num [1:83] NA 1.8 2.4 2.3 0.7 2.2 1.4 NA 2.9 NA ...
##  $ sleep_cycle : num [1:83] NA NA NA 0.133 0.667 ...
##  $ awake       : num [1:83] 11.9 7 9.6 9.1 20 9.6 15.3 17 13.9 21 ...
##  $ brainwt     : num [1:83] NA 0.0155 NA 0.00029 0.423 NA NA NA 0.07 0.098
2 ...
##  $ bodywt      : num [1:83] 50 0.48 1.35 0.019 600 ...
```

```
glimpse(raw)
```

```
## Rows: 83
## Columns: 11
## $ name         <chr> "Cheetah", "Owl monkey", "Mountain beaver", "Greater
shor…
## $ genus        <chr> "Acinonyx", "Aotus", "Aplodontia", "Blarina", "Bos",
"Bra…
## $ vore         <chr> "carni", "omni", "herbi", "omni", "herbi", "herbi", "
```

```
carn…
## $ order        <chr> "Carnivora", "Primates", "Rodentia", "Soricomorpha",
"Art…
## $ conservation <chr> "lc", NA, "nt", "lc", "domesticated", NA, "vu", NA, "
dome…
## $ sleep_total  <dbl> 12.1, 17.0, 14.4, 14.9, 4.0, 14.4, 8.7, 7.0, 10.1, 3.
0, 5…
## $ sleep_rem    <dbl> NA, 1.8, 2.4, 2.3, 0.7, 2.2, 1.4, NA, 2.9, NA, 0.6, 0.
8, …
## $ sleep_cycle  <dbl> NA, NA, NA, 0.1333333, 0.6666667, 0.7666667, 0.383333
3, N…
## $ awake        <dbl> 11.9, 7.0, 9.6, 9.1, 20.0, 9.6, 15.3, 17.0, 13.9, 21.
0, 1…
## $ brainwt      <dbl> NA, 0.01550, NA, 0.00029, 0.42300, NA, NA, NA, 0.0700
0, 0…
## $ bodywt       <dbl> 50.000, 0.480, 1.350, 0.019, 600.000, 3.850, 20.490,
0.04…
```

3.2 *Summary statistics of the raw data*

```
summary(raw)
```

```
##         name           genus           vore           order       conservati
on
##  Length   :83   Length   :83   Length   :83   Length   :83   Length   :83
##  N.unique :83   N.unique :77   N.unique : 4   N.unique :19   N.unique : 6
##  N.blank  : 0   N.blank  : 0   N.blank  : 0   N.blank  : 0   N.blank  : 0
##  Min.nchar: 3   Min.nchar: 3   Min.nchar: 4   Min.nchar: 6   Min.nchar: 2
##  Max.nchar:30   Max.nchar:13   Max.nchar: 7   Max.nchar:15   Max.nchar:12
##                                NAs      : 7                  NAs      :29
##

##   sleep_total      sleep_rem      sleep_cycle         awake
##  Min.   : 1.90   Min.   :0.100   Min.   :0.1167   Min.   : 4.10
##  1st Qu.: 7.85   1st Qu.:0.900   1st Qu.:0.1833   1st Qu.:10.25
##  Median :10.10   Median :1.500   Median :0.3333   Median :13.90
##  Mean   :10.43   Mean   :1.875   Mean   :0.4396   Mean   :13.57
##  3rd Qu.:13.75   3rd Qu.:2.400   3rd Qu.:0.5792   3rd Qu.:16.15
##  Max.   :19.90   Max.   :6.600   Max.   :1.5000   Max.   :22.10
##                  NAs    :22      NAs    :51
##     brainwt            bodywt
##  Min.   :0.00014   Min.   :   0.005
##  1st Qu.:0.00290   1st Qu.:   0.174
##  Median :0.01240   Median :   1.670
```

```
##  Mean   :0.28158   Mean   : 166.136
##  3rd Qu.:0.12550   3rd Qu.:  41.750
##  Max.   :5.71200   Max.   :6654.000
##  NAs    :27
```

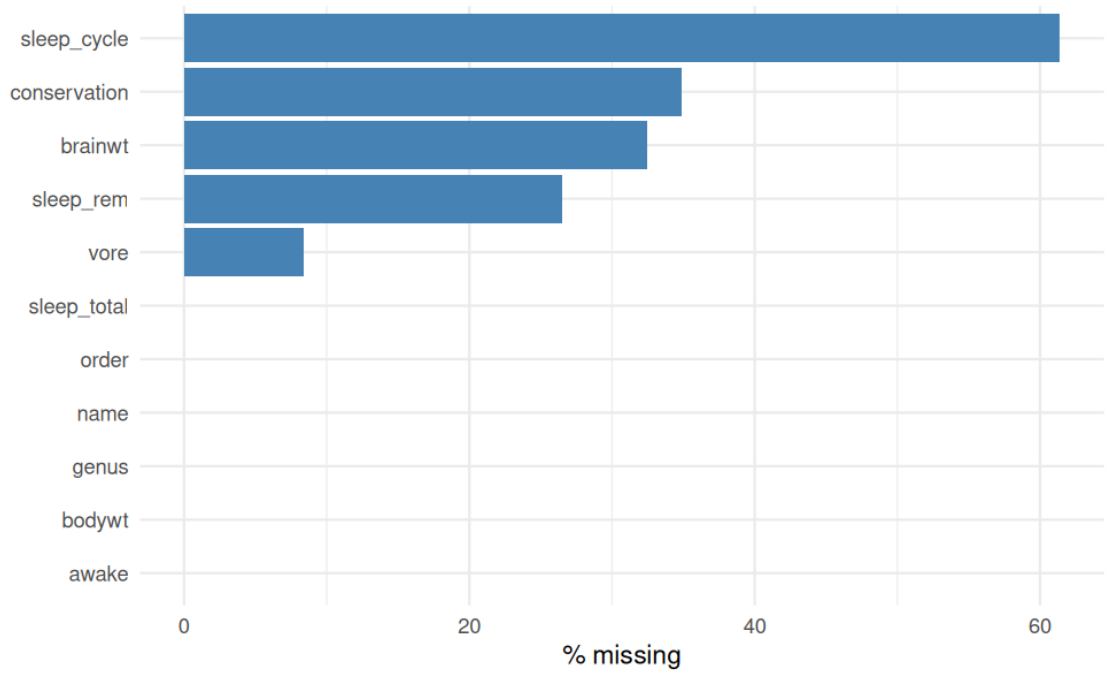
3.3 *Missing values*

```
miss_tbl <- tibble(
  variable    = names(raw),
  n_missing   = colSums(is.na(raw)),
  pct_missing = round(100 * n_missing / nrow(raw), 1)
) %>% arrange(desc(n_missing))
kable(miss_tbl, caption = "Missing values per variable in the raw data")
```

Missing values per variable in the raw data

variable	n_missing	pct_missing
sleep_cycle	51	61.4
conservation	29	34.9
brainwt	27	32.5
sleep_rem	22	26.5
vore	7	8.4
name	0	0.0
genus	0	0.0
order	0	0.0
sleep_total	0	0.0
awake	0	0.0
bodywt	0	0.0

```
ggplot(miss_tbl, aes(x = reorder(variable, pct_missing), y = pct_missing)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(x = NULL, y = "% missing")
```



Percentage of missing values per variable (raw data)

3.4 *Duplicates*

```
cat("Duplicated rows:", sum(duplicated(raw)), "\n")
```

```
## Duplicated rows: 0
```

```
cat("Duplicated animal names:", sum(duplicated(raw$name)), "\n")
```

```
## Duplicated animal names: 0
```

4. Data Cleaning and Transformation

4.1 *Type correction and text standardisation*

Leading/trailing whitespace is trimmed from text columns, and categorical columns are converted to factors so that R treats them as discrete groups.

```
df <- raw %>%
  mutate(across(where(is.character), str_trim)) %>%
  mutate(across(c(order, genus, vore, conservation), as.factor))
str(df)
```

```
## tibble [83 × 11] (S3: tbl_df/tbl/data.frame)
##  $ name         : chr [1:83] "Cheetah" "Owl monkey" "Mountain beaver" "Grea
ter short-tailed shrew" ...
##  $ genus        : Factor w/ 77 levels "Acinonyx","Aotus",..: 1 2 3 4 5 6 7
8 9 10 ...
##  $ vore         : Factor w/ 4 levels "carni","herbi",..: 1 4 2 4 2 2 1 NA 1
```

```
 2 ...
##  $ order       : Factor w/ 19 levels "Afrosoricida",..: 3 15 17 19 2 14 3
17 3 2 ...
##  $ conservation: Factor w/ 6 levels "cd","domesticated",..: 4 NA 5 4 2 NA
6 NA 2 4 ...
##  $ sleep_total : num [1:83] 12.1 17 14.4 14.9 4 14.4 8.7 7 10.1 3 ...
##  $ sleep_rem   : num [1:83] NA 1.8 2.4 2.3 0.7 2.2 1.4 NA 2.9 NA ...
##  $ sleep_cycle : num [1:83] NA NA NA 0.133 0.667 ...
##  $ awake       : num [1:83] 11.9 7 9.6 9.1 20 9.6 15.3 17 13.9 21 ...
##  $ brainwt     : num [1:83] NA 0.0155 NA 0.00029 0.423 NA NA NA 0.07 0.098
2 ...
##  $ bodywt      : num [1:83] 50 0.48 1.35 0.019 600 ...
```

4.2 *Handling missing values*

The strategy depends on the variable type and how much is missing:

Situation	Strategy	Reason
Column with more than 50% missing	Drop the column	Imputing most of a column would mostly invent data
Missing categorical value	Recode as an explicit "unknown" level	Missingness may itself be informative; avoids deleting rows
Missing numeric value	Median of the animal's diet group (vore), falling back to the overall median	Median is robust to the heavy skew seen in brain weight; group median keeps biological structure
Imputed numeric values	Add a logical flag column	Keeps imputation transparent for later modelling

```r
# (a) Drop columns with too much missingness
threshold <- 0.5
high_miss <- names(which(colMeans(is.na(df)) > threshold))
cat("Columns dropped (>", threshold * 100, "% missing):", high_miss, "\n")

## Columns dropped (> 50 % missing): sleep_cycle

df <- df %>% select(-all_of(high_miss))

# (b) Categorical NA -> explicit "unknown" level
df <- df %>%
  mutate(across(c(vore, conservation),
                ~ factor(replace_na(as.character(.x), "unknown"))))

# (c) Numeric NA -> group median with overall-median fallback
```

```
impute_group_median <- function(x, g) {
  out <- ave(x, g, FUN = function(v) { v[is.na(v)] <- median(v, na.rm = TRUE);
 v })
  out[is.na(out)] <- median(x, na.rm = TRUE)
  out
}

df <- df %>%
  mutate(brainwt_imputed  = is.na(brainwt),
         sleep_rem_imputed = is.na(sleep_rem),
         brainwt  = impute_group_median(brainwt,  vore),
         sleep_rem = impute_group_median(sleep_rem, vore))

# Verification: no missing values should remain
kable(tibble(variable = names(df), n_missing = colSums(is.na(df))) %>%
        filter(n_missing > 0),
      caption = "Remaining missing values (empty means none)")
```

Remaining missing values (empty means none)

variable n_missing

```
cat("Total remaining NA:", sum(is.na(df)), "\n")

## Total remaining NA: 0
```

Sensitivity check: *imputation should not distort the distributions much.*

```
check <- tibble(
  variable    = c("sleep_rem", "brainwt"),
  mean_before = c(mean(raw$sleep_rem, na.rm = TRUE), mean(raw$brainwt, na.rm
= TRUE)),
  mean_after  = c(mean(df$sleep_rem), mean(df$brainwt)),
  sd_before   = c(sd(raw$sleep_rem, na.rm = TRUE), sd(raw$brainwt, na.rm = TR
UE)),
  sd_after    = c(sd(df$sleep_rem), sd(df$brainwt))
) %>% mutate(across(where(is.numeric), ~ round(.x, 4)))
kable(check, caption = "Mean and SD before and after imputation")
```

Mean and SD before and after imputation

variable	mean_before	mean_after	sd_before	sd_after
sleep_rem	1.8754	1.8102	1.2983	1.1528
brainwt	0.2816	0.1974	0.9764	0.8090

Imputing brain weight with the group median lowered its mean (0.2816 to 0.1974) and its spread (SD 0.9764 to 0.8090), because the original values are heavily skewed by a few very large animals. This is a limitation of simple imputation, and the log transformation in Section 4.4 reduces its effect on later analysis. The mean of sleep_rem changed much less.

4.3 *Outlier detection*

Outliers are detected with the **IQR rule**: a value is flagged if it lies below Q1 − 1.5·IQR or above Q3 + 1.5·IQR.

```
iqr_outliers <- function(x) {
  q   <- quantile(x, c(0.25, 0.75), na.rm = TRUE)
  iqr <- q[2] - q[1]
  x < q[1] - 1.5 * iqr | x > q[2] + 1.5 * iqr
}

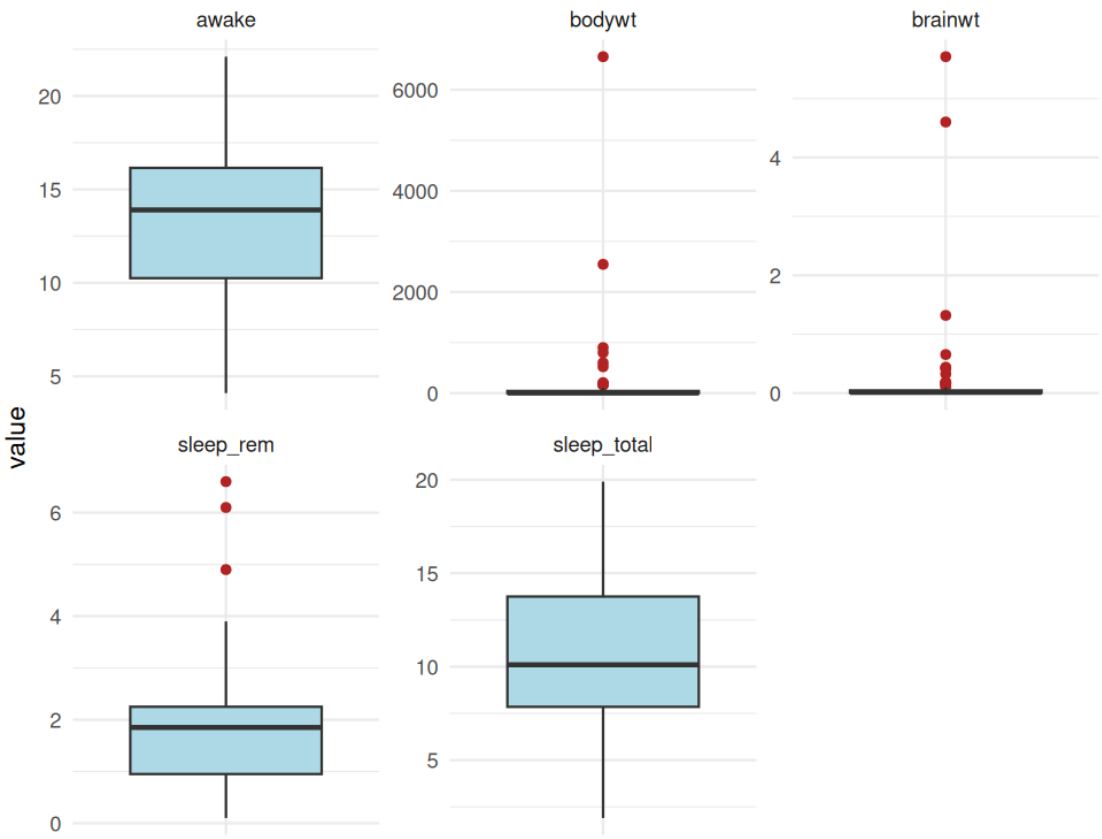
num_vars <- df %>% select(where(is.numeric)) %>% names()

outlier_summary <- map_dfr(num_vars, function(v) {
  flag <- iqr_outliers(df[[v]])
  tibble(variable = v, n_outliers = sum(flag), pct_outliers = round(100 * mea
n(flag), 1))
})
kable(outlier_summary, caption = "Outliers per numeric variable (IQR rule)")
```

Outliers per numeric variable (IQR rule)

variable	n_outliers	pct_outliers
sleep_total	0	0.0
sleep_rem	3	3.6
awake	0	0.0
brainwt	16	19.3
bodywt	11	13.3

```
df %>%
  select(all_of(num_vars)) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "value") %>%
  ggplot(aes(x = variable, y = value)) +
  geom_boxplot(fill = "lightblue", outlier.colour = "firebrick") +
  facet_wrap(~ variable, scales = "free") +
  theme(axis.text.x = element_blank(), axis.title.x = element_blank())
```



Boxplots of numeric variables (free scales) showing outliers

```
df %>%
  filter(iqr_outliers(bodywt)) %>%
  select(name, bodywt, brainwt) %>%
  arrange(desc(bodywt)) %>%
  kable(caption = "Body-weight outliers")
```

Body-weight outliers

name	bodywt	brainwt
African elephant	6654.000	5.712000
Asian elephant	2547.000	4.603000
Giraffe	899.995	0.012285
Pilot whale	800.000	0.044500
Cow	600.000	0.423000
Horse	521.000	0.655000
Brazilian tapir	207.501	0.169000
Donkey	187.000	0.419000
Bottle-nosed dolphin	173.330	0.044500

name	bodywt	brainwt
Tiger	162.564	0.044500
Lion	161.499	0.044500

Decision: *These outliers are genuine biological measurements (e.g. large mammals), not data-entry errors, so deleting them would remove real information. Instead they are flagged and their influence is reduced with a log transformation* (Section 4.4).

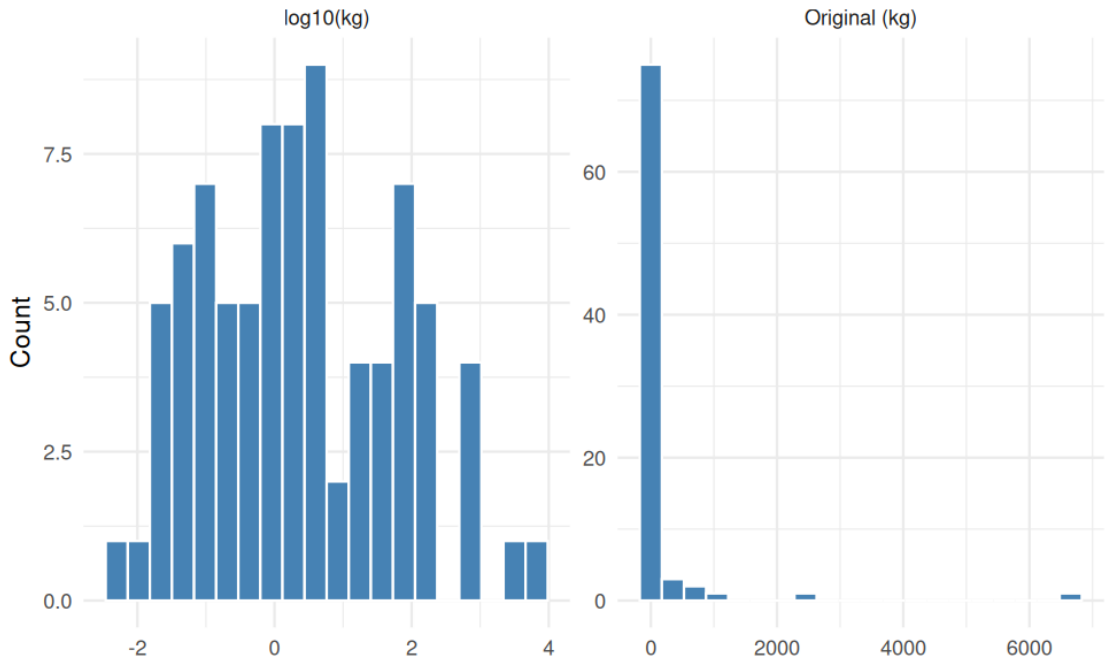
```
df <- df %>% mutate(bodywt_outlier = iqr_outliers(bodywt),
                    brainwt_outlier = iqr_outliers(brainwt))
```

4.4 *Transformation and normalisation*

Log transformation (to fix skew)

```
df <- df %>% mutate(log_bodywt  = log10(bodywt),
                    log_brainwt = log10(brainwt))

bind_rows(tibble(scale = "Original (kg)", value = df$bodywt),
          tibble(scale = "log10(kg)",     value = df$log_bodywt)) %>%
  ggplot(aes(value)) +
  geom_histogram(bins = 20, fill = "steelblue", colour = "white") +
  facet_wrap(~ scale, scales = "free") +
  labs(x = NULL, y = "Count")
```



Body weight before and after log10 transformation

Z-score and min-max scaling

Z-score standardisation (mean 0, SD 1) suits methods sensitive to variance; min-max scaling (0 to 1) suits methods sensitive to range.

```
minmax <- function(x) (x - min(x)) / (max(x) - min(x))
scale_vars <- c("sleep_total", "sleep_rem", "awake", "log_brainwt", "log_body
wt")

df_norm <- df %>%
  mutate(across(all_of(scale_vars),
                list(z = ~ as.numeric(scale(.x)), mm = minmax)))

z_check <- sapply(select(df_norm, ends_with("_z")),
                  function(x) c(mean = mean(x), sd = sd(x)))
kable(round(z_check, 3), caption = "Z-scored variables: mean ~ 0, SD = 1")
```

Z-scored variables: mean ~ 0, SD = 1

	sleep_total_z	sleep_rem_z	awake_z	log_brainwt_z	log_bodywt_z
mean	0	0	0	0	0
sd	1	1	1	1	1

```
mm_check <- sapply(select(df_norm, ends_with("_mm")),
                   function(x) c(min = min(x), max = max(x)))
kable(round(mm_check, 3), caption = "Min-max scaled variables: range 0 to 1")
```

Min-max scaled variables: range 0 to 1

	sleep_total_mm	sleep_rem_mm	awake_mm	log_brainwt_mm	log_bodywt_mm
min	0	0	0	0	0
max	1	1	1	1	1

4.5 *Encoding categorical variables*

Three different encodings are used, each matched to the variable:

Variable	Encoding	Reason
vore (5 levels, no order)	One-hot (dummy)	No natural order between diets
conservation	Ordinal	Threat categories have a natural order of risk
order (many levels)	Frequency encoding	One-hot would add too

Variable	Encoding	Reason
		many columns for 83 rows

```
# One-hot encoding for vore
onehot <- model.matrix(~ vore - 1, data = df_norm)
colnames(onehot) <- paste0("vore_", levels(df_norm$vore))

# Ordinal encoding for conservation (least to most threatened).
# "domesticated" and "unknown" are not points on the threat scale -> NA.
threat_order <- c("lc", "cd", "nt", "vu", "en")

df_enc <- bind_cols(df_norm, as_tibble(onehot)) %>%
  mutate(conservation_ord = match(as.character(conservation), threat_order),
         order_freq = ave(rep(1, n()), order, FUN = sum) / n())

df_enc %>%
  select(name, vore, starts_with("vore_"), conservation, conservation_ord,
         order, order_freq) %>%
  head(8) %>%
  kable(digits = 3, caption = "First rows of the encoded variables")
```

First rows of the encoded variables

name	vore	vore_carni	vore_herbi	vore_insecti	vore_omni	vore_unknown	conservation	conservation_ord	order	order_freq
Cheetah	carni	1	0	0	0	0	lc	1	Carnivora	0.145
Owl monkey	omni	0	0	0	1	0	unknown	NA	Primates	0.145
Mountain beaver	herbi	0	1	0	0	0	nt	3	Rodentia	0.265
Greater short-tailed shrew	omni	0	0	0	1	0	lc	1	Soricomorpha	0.060
Cow	herbi	0	1	0	0	0	domesticated	NA	Artiodactyla	0.072

name	vore	vore_carni	vore_herbi	vore_insecti	vore_omni	vore_unknown	conservation	conservation_ord	order	order_freq
Three-toed sloth	herbi	0	1	0	0	0	unknown	NA	Pilosa	0.012
Northern fur seal	carni	1	0	0	0	0	vu	4	Carnivora	0.145
Vesper mouse	unknown	0	0	0	0	1	unknown	NA	Rodentia	0.265

4.6 *Cleaned dataset*

```
dim(df_enc)

## [1] 83 33

write_csv(df_enc, "msleep_clean.csv")
```

5. Exploratory Data Analysis

5.1 *Structure and summary of the cleaned data*

```
str(df_enc %>% select(name, order, vore, conservation, all_of(num_vars)))

## tibble [83 × 9] (S3: tbl_df/tbl/data.frame)
##  $ name        : chr [1:83] "Cheetah" "Owl monkey" "Mountain beaver" "Grea
ter short-tailed shrew" ...
##  $ order       : Factor w/ 19 levels "Afrosoricida",..: 3 15 17 19 2 14 3
17 3 2 ...
##  $ vore        : Factor w/ 5 levels "carni","herbi",..: 1 4 2 4 2 2 1 5 1
2 ...
##  $ conservation: Factor w/ 7 levels "cd","domesticated",..: 4 6 5 4 2 6 7
6 2 4 ...
##  $ sleep_total : num [1:83] 12.1 17 14.4 14.9 4 14.4 8.7 7 10.1 3 ...
##  $ sleep_rem   : num [1:83] 1.95 1.8 2.4 2.3 0.7 2.2 1.4 2 2.9 0.95 ...
##  $ awake       : num [1:83] 11.9 7 9.6 9.1 20 9.6 15.3 17 13.9 21 ...
##  $ brainwt     : num [1:83] 0.0445 0.0155 0.01229 0.00029 0.423 ...
##  $ bodywt      : num [1:83] 50 0.48 1.35 0.019 600 ...

summary(df_enc %>% select(vore, conservation, all_of(num_vars)))
```

```
##       vore          conservation  sleep_total      sleep_rem        awake
##  carni  :19   cd          : 2    Min.   : 1.90   Min.   :0.10   Min.   : 4.
10
##  herbi  :32   domesticated:10    1st Qu.: 7.85   1st Qu.:0.95   1st Qu.:10.
25
##  insecti: 5   en          : 4    Median :10.10   Median :1.85   Median :13.
90
##  omni   :20   lc          :27    Mean   :10.43   Mean   :1.81   Mean   :13.
57
##  unknown: 7   nt          : 4    3rd Qu.:13.75   3rd Qu.:2.25   3rd Qu.:16.
15
##               unknown     :29    Max.   :19.90   Max.   :6.60   Max.   :22.
10
##               vu          : 7

##     brainwt            bodywt
##  Min.   :0.00014   Min.   :   0.005
##  1st Qu.:0.00560   1st Qu.:   0.174
##  Median :0.01229   Median :   1.670
##  Mean   :0.19743   Mean   : 166.136
##  3rd Qu.:0.04450   3rd Qu.:  41.750
##  Max.   :5.71200   Max.   :6654.000
##
```

5.2 *Descriptive statistics*

```
desc <- df %>%
  select(all_of(num_vars)) %>%
  pivot_longer(everything(), names_to = "variable") %>%
  group_by(variable) %>%
  summarise(n = n(), mean = mean(value), median = median(value), sd = sd(valu
e),
            min = min(value), max = max(value),
            skewness = mean((value - mean(value))^3) / sd(value)^3,
            .groups = "drop") %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)))
kable(desc, caption = "Descriptive statistics of numeric variables")
```

Descriptive statistics of numeric variables

variable	n	mean	median	sd	min	max	skewness
awake	83	13.567	13.900	4.452	4.100	22.100	-0.051
bodywt	83	166.136	1.670	786.840	0.005	6654.000	7.100
brainwt	83	0.197	0.012	0.809	0.000	5.712	5.789
sleep_rem	83	1.810	1.850	1.153	0.100	6.600	1.663
sleep_total	83	10.434	10.100	4.450	1.900	19.900	0.052

```
by_vore <- df %>%
  group_by(vore) %>%
```

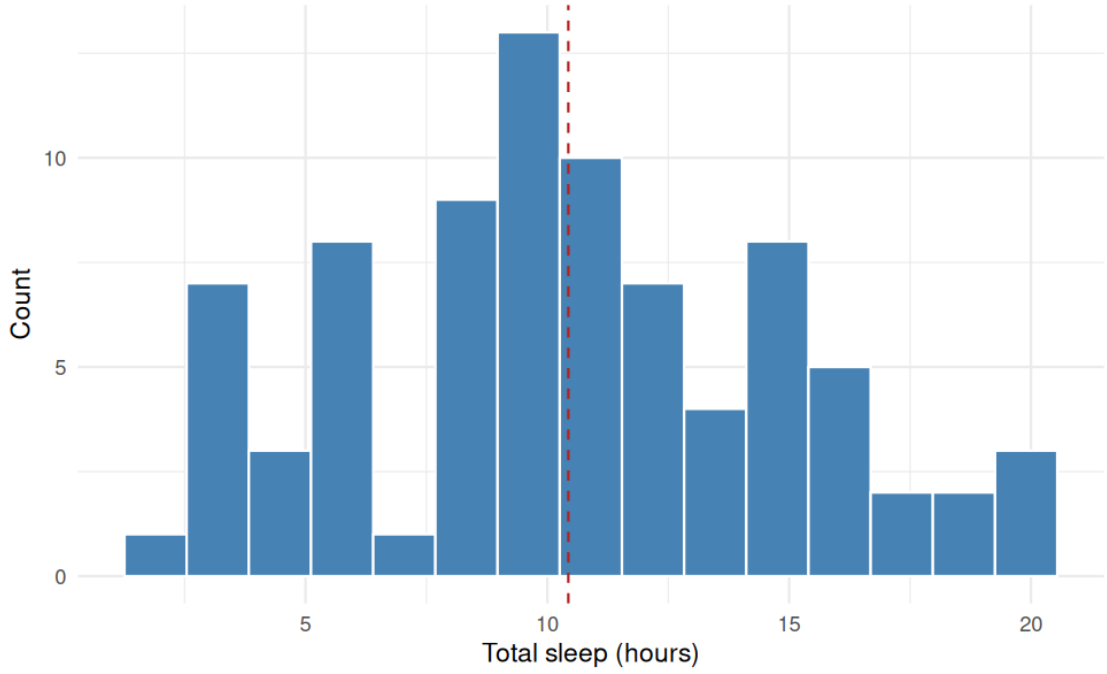
```
  summarise(n = n(),
            mean_sleep   = round(mean(sleep_total), 2),
            median_sleep = round(median(sleep_total), 2),
            mean_rem     = round(mean(sleep_rem), 2),
            .groups = "drop") %>%
  arrange(desc(mean_sleep))
kable(by_vore, caption = "Sleep by diet group")
```

Sleep by diet group

vore	n	mean_sleep	median_sleep	mean_rem
insecti	5	14.94	18.1	3.42
omni	20	10.93	9.9	1.95
carni	19	10.38	10.4	2.13
unknown	7	10.19	10.6	1.91
herbi	32	9.51	10.3	1.26

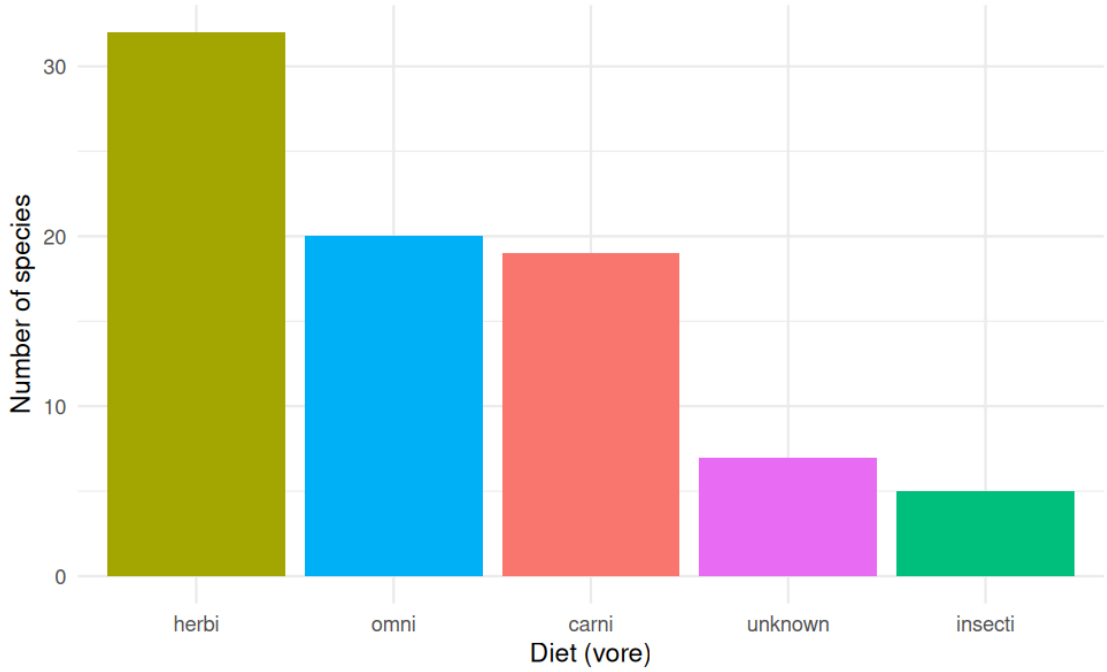
5.3 *Distributions*

```
ggplot(df, aes(sleep_total)) +
  geom_histogram(bins = 15, fill = "steelblue", colour = "white") +
  geom_vline(aes(xintercept = mean(sleep_total)), colour = "firebrick", linetype = "dashed") +
  labs(x = "Total sleep (hours)", y = "Count")
```



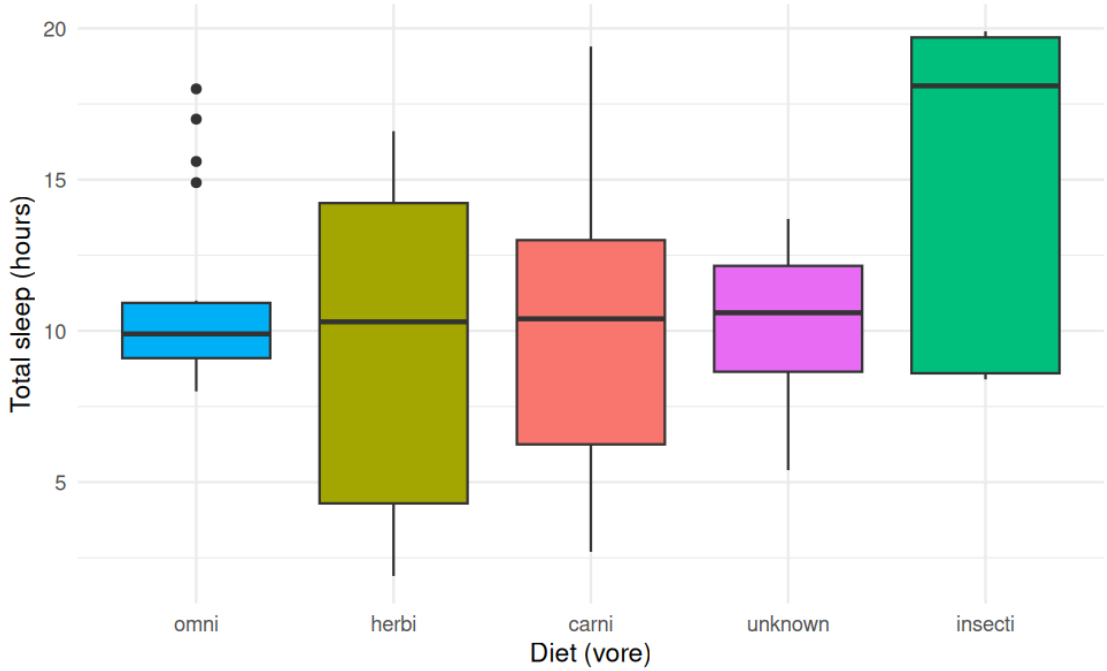
Distribution of total sleep (hours per day)

```r
ggplot(df, aes(x = fct_infreq(vore), fill = vore)) +
  geom_bar(show.legend = FALSE) +
  labs(x = "Diet (vore)", y = "Number of species")
```



Number of species by diet

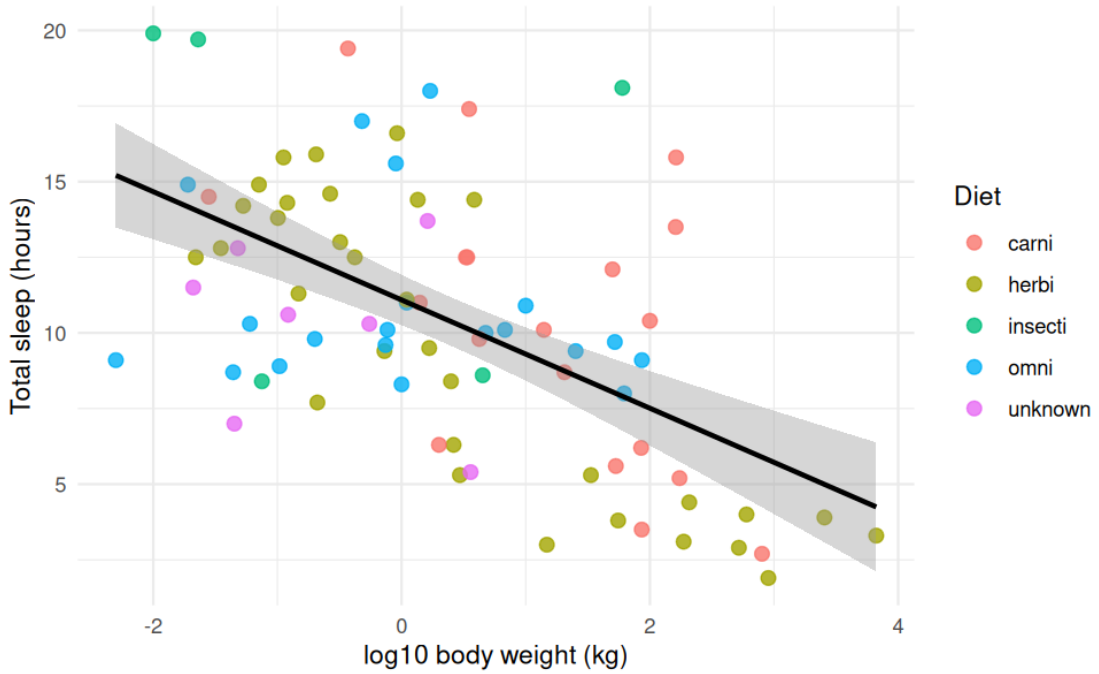
```r
ggplot(df, aes(x = reorder(vore, sleep_total, median), y = sleep_total, fill
= vore)) +
  geom_boxplot(show.legend = FALSE) +
  labs(x = "Diet (vore)", y = "Total sleep (hours)")
```



Total sleep by diet group

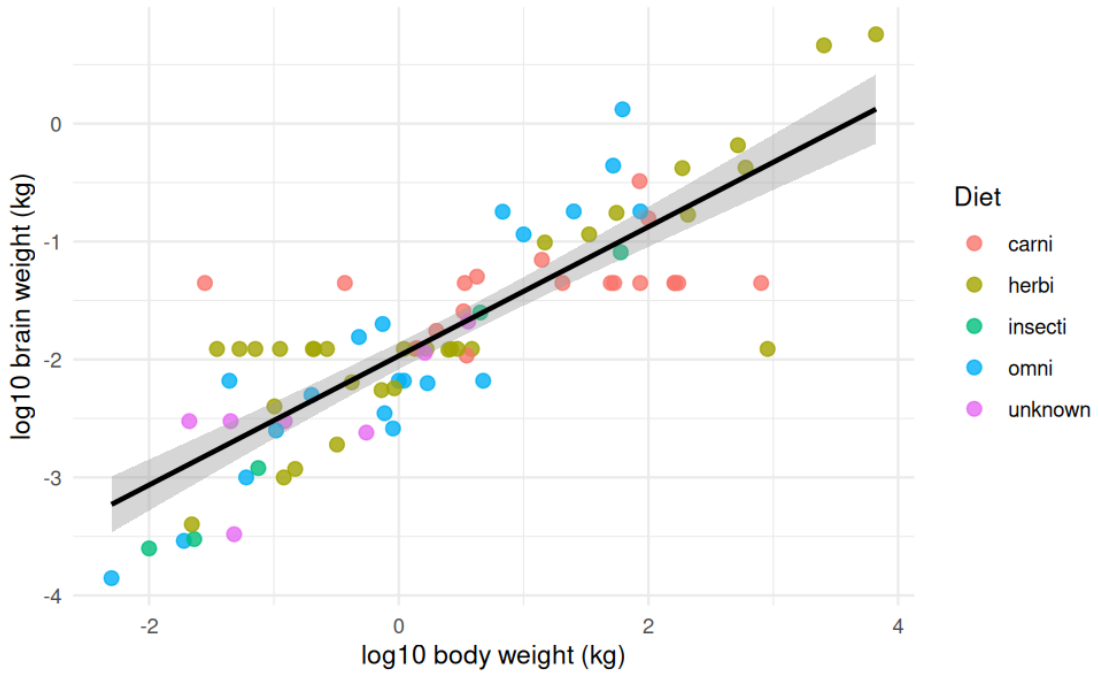
5.4 *Relationships between variables*

```
ggplot(df, aes(log_bodywt, sleep_total)) +
  geom_point(aes(colour = vore), size = 2.5, alpha = 0.8) +
  geom_smooth(method = "lm", formula = y ~ x, colour = "black") +
  labs(x = "log10 body weight (kg)", y = "Total sleep (hours)", colour = "Die
t")
```



Total sleep versus log10 body weight, coloured by diet

```
ggplot(df, aes(log_bodywt, log_brainwt)) +
  geom_point(aes(colour = vore), size = 2.5, alpha = 0.8) +
  geom_smooth(method = "lm", formula = y ~ x, colour = "black") +
  labs(x = "log10 body weight (kg)", y = "log10 brain weight (kg)", colour =
"Diet")
```



log10 brain weight versus log10 body weight

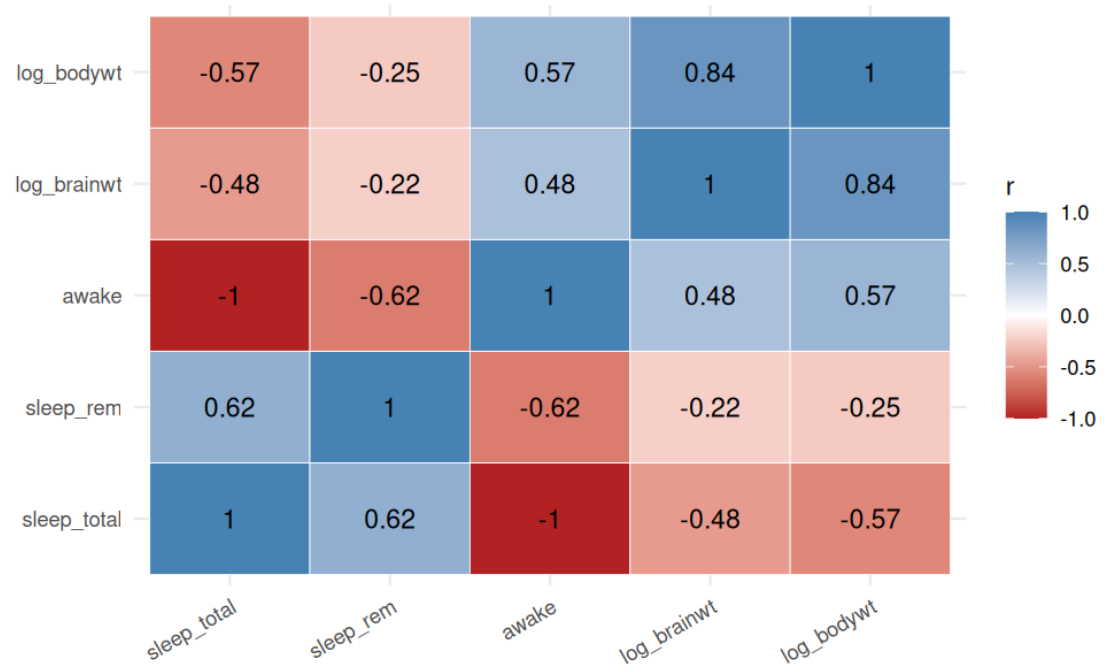
5.5 *Correlation analysis*

```
cor_vars <- c("sleep_total", "sleep_rem", "awake", "log_brainwt", "log_bodywt
")
cmat <- cor(df[cor_vars])
kable(round(cmat, 2), caption = "Pearson correlation matrix")
```

Pearson correlation matrix

	sleep_total	sleep_rem	awake	log_brainwt	log_bodywt
sleep_total	1.00	0.62	-1.00	-0.48	-0.57
sleep_rem	0.62	1.00	-0.62	-0.22	-0.25
awake	-1.00	-0.62	1.00	0.48	0.57
log_brainwt	-0.48	-0.22	0.48	1.00	0.84
log_bodywt	-0.57	-0.25	0.57	0.84	1.00

```
as.data.frame(as.table(cmat)) %>%
  ggplot(aes(Var1, Var2, fill = Freq)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = round(Freq, 2))) +
  scale_fill_gradient2(low = "firebrick", mid = "white", high = "steelblue",
                       limits = c(-1, 1)) +
  labs(x = NULL, y = NULL, fill = "r") +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```



Correlation heatmap

```
top_pairs <- as.data.frame(as.table(cmat)) %>%
  filter(as.character(Var1) < as.character(Var2)) %>%
```

```
  mutate(Freq = round(Freq, 3)) %>%
  arrange(desc(abs(Freq))) %>%
  rename(Variable_1 = Var1, Variable_2 = Var2, r = Freq)
kable(top_pairs, caption = "Variable pairs ranked by absolute correlation")
```

Variable pairs ranked by absolute correlation

Variable_1	Variable_2	r
awake	sleep_total	-1.000
log_bodywt	log_brainwt	0.838
sleep_rem	sleep_total	0.623
awake	sleep_rem	-0.623
log_bodywt	sleep_total	-0.569
awake	log_bodywt	0.569
log_brainwt	sleep_total	-0.477
awake	log_brainwt	0.477
log_bodywt	sleep_rem	-0.251
log_brainwt	sleep_rem	-0.225

6. Key Insights

- **Data quality:-** The raw data had 136 missing cells. After cleaning 0 remain in the analysis columns.
- **Skewness:-** Body and brain weight are strongly right-skewed on the raw scale; the log10 transform makes them far more symmetric and suitable for correlation and regression.
- **Sleep and body size:-** The correlation between sleep and log body weight is -0.57, a negative relationship: larger animals tend to sleep less.
- **Brain and body size:-** Log brain weight and log body weight are highly correlated (r = 0.84) reflecting scaling.
- **Sleep and REM:-** Total sleep and REM sleep have r = 0.62.
- **Redundancy:-** Awake equals 24 minus sleep_total so the two are negatively correlated (r = -1); only one should be used in a model.
- **Diet:-** The diet group with the highest average sleep is insecti. Group sizes are uneven (only 5 species are insectivores), so this should be treated as a preliminary observation, not a conclusion.

7. Conclusion and Limitations

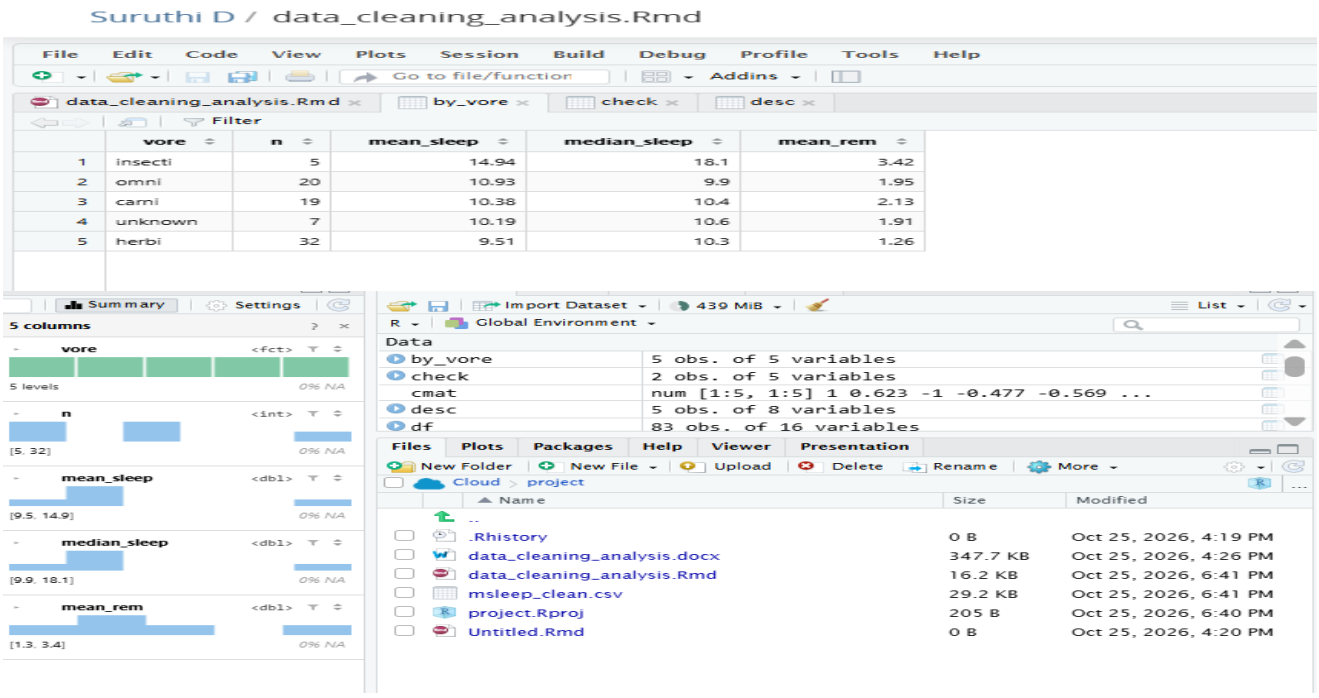
The dataset was cleaned through column removal, explicit-unknown recoding and group-median imputation; outliers were detected with the IQR rule and retained after log transformation; numeric variables were standardised and min-max scaled; and categorical variables were encoded with one-hot, ordinal and frequency methods.

Limitations:

- With only 83 species, group-level results (especially small diet groups) are unstable.
- Median imputation shrinks variance slightly; more advanced methods (e.g. multiple imputation with mice) could be tested.
- Correlation does not imply causation, and species are not independent observations because related species share ancestry.

Possible next steps: *Regression of sleep on body size and diet, clustering of species, and multiple imputation.*

8. Screenshots of the RStudio Workflow



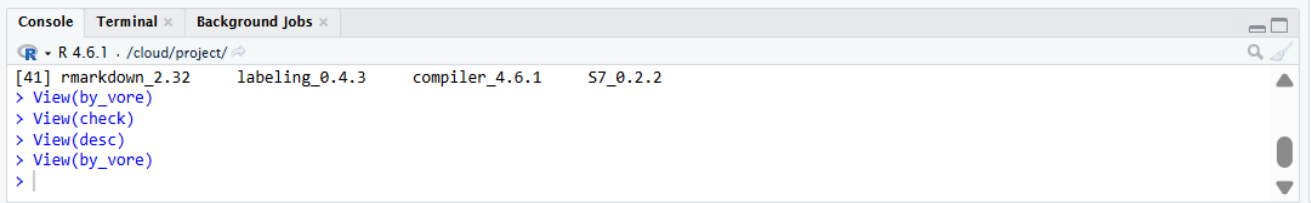


Figure: RStudio workspace showing the analysis script, result tables and output files

Appendix: Reproducibility

```
sessionInfo()

## R version 4.6.1 (2026-06-24)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 24.04.5 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;
  LAPACK version 3.12.0
##
## locale:
##  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
##  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
##  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
## [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
##
## time zone: UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] knitr_1.52      lubridate_1.9.5 forcats_1.0.1   stringr_1.6.0
##  [5] dplyr_1.2.1     purrr_1.2.2     readr_2.2.0     tidyr_1.3.2
##  [9] tibble_3.3.1    ggplot2_4.0.3   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] generics_0.1.4     stringi_1.8.9      lattice_0.22-9     hms_1.1.4

##  [5] digest_0.6.39      magrittr_2.0.5     evaluate_1.0.5     grid_4.6.1

##  [9] timechange_0.4.0   RColorBrewer_1.1-3 fastmap_1.2.0      Matrix_1.7-5

## [13] mgcv_1.9-4         scales_1.4.0       cli_3.6.6          rlang_1.3.0

## [17] crayon_1.5.3       bit64_4.8.6        splines_4.6.1      withr_3.0.3
```

```
## [21] yaml_2.3.12        otel_0.2.0         tools_4.6.1        parallel_4.6.
1
## [25] tzdb_0.5.0         vctrs_0.7.3        R6_2.6.1           lifecycle_1.
0.5
## [29] bit_4.6.0          vroom_1.7.1        pkgconfig_2.0.3    pillar_1.11.
1
## [33] gtable_0.3.6       glue_1.8.1         xfun_0.62          tidyselect_1.
2.1
## [37] rstudioapi_0.19.0  farver_2.1.2       htmltools_0.5.9    nlme_3.1-169

## [41] rmarkdown_2.32     labeling_0.4.3     compiler_4.6.1     S7_0.2.2
```